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Participatory Impact Assessment of Soil and Water Conservation Scenarios in Oum Zessar Watershed, Tunisia

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Abstract Environmental threats and progressive degradation of natural resources are considered critical impediments to sustainable development. This paper reports on a participatory impact assessment of alternative soil and water conservation (SWC) scenarios in the Oum Zessar watershed, Tunisia. The first objective was to assess the impact of three SWC scenarios on key social, economic and environmental land use functions. The second objective was to test and evaluate the applicability of the 'Framework for Participatory Impact Assessment (FoPIA)' for assessing scenario impacts in the context of a developing country, in this case Tunisia. The assessed scenarios included: the originally planned SWC policy implementation at 85 % coverage of arable land of the watershed, the current implementation (70 %), and a hypothetical expansion of SWC measures to the entire watershed (100 %). Our results suggest that implementation of the SWC policy at 100 % coverage of arable land achieves the maximum socioeconomic benefit. However, if stakeholders' preferences regarding land use functions are taken into account, and considering the fact that the implementation of SWC measures also implies some negative changes to traditional landscapes and the natural system, SWC implementation at 85 % coverage of arable land might be preferable. The FoPIA approved to be a useful tool for conducting a holistic sustainability impact assessment of SWC scenarios and for studying the most intriguing sustainability problems while providing possible recommendations towards sustainable development. We conclude that participatory impact assessment contributes to an enhanced regional understanding of key linkages between policy effects and sustainable development, which provides the foundation for improved policy decision making.

Keywords Sustainability impact assessment · Scenario study · FoPIA · Land management · Land use functions · Stakeholder participation

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Introduction

Environmental threats and the progressive degradation of natural resources pose critical impediments to sustainable development, particularly in the arid regions of Tunisia. The Oum Zessar watershed in the Médenine region in southeastern Tunisia faces severe land degradation problems due to limited water resources, growing population, increasing land competition and agricultural intensification (Ouessar and others 2009; Schiettecatte and others 2005).

In response to ongoing degradation problems, the Tunisian government has invested 562 million Dinars (=US $389 mil)

since 1990 on soil and water conservation (SWC) measures at the national level (Sghaier and others 2009). The major goals of the SWC policy are to tackle land degradation in a vulnerable environment and to enhance the capacity of the land for agricultural production through the promotion of both traditional and modern water-harvesting techniques (Ouessar and others 2009). Policy initiatives have mainly been realised through subsidies provided by the government. These have included regional infrastructure projects, subsidies and financial support for farmers for constructing and maintaining water harvesting facilities at the local level (Fleskens and others 2005; Ouessar and others 2004; Schiettecatte and others 2005). Large parts of the Oum Zessar watershed were enrolled in the SWC policy program and have since experienced dynamic changes in land management due to the reorientation of agricultural production from small-scale and subsistence farming towards market-oriented agriculture.

Regional decision makers are currently evaluating the 'success of the SWC policy' and considering a possible expansion of the SWC policy in the Oum Zessar watershed. To ensure policy efficiency, decision makers demand a comprehensive and reliable assessment of the possible impacts of policy changes on the economic, environmental and social components of development (O'Farrell and Anderson 2010; Pope and Grace 2006; Scrieciu 2007). An impact assessment of the SWC policy towards sustainable development could support the decision-making process. In this study, we demonstrate the use of the Framework for Participatory Impact Assessment (FoPIA) by conducting a case study-based sustainability impact assessment of SWC scenarios in Oum Zessar watershed. For this purpose, the FoPIA, which was originally developed for the European context (Morris and others 2011), was adapted to the regional conditions in Oum Zessar watershed.

Sustainability impact assessment (SIA) is an increasingly accepted way of incorporating this holistic perspective into policy assessments (Boulanger and Brechet 2005; De Ridder and others 2007; George and Kirkpatrick 2006). Currently, a wide range of methods and tools are available in the field of SIA including both analytical (i.e., computer-based models) and more qualitative methods (Hacking and Guthrie 2002; Rotmans 1998). Although there is no common procedure for SIA, most approaches characterise sustainability in terms of indicators (Binder and others 2010; Diaz-Balteiro and Romero 2004; Paracchini and others 2011; Walter and Stutzel 2009; Wiggering and others 2006). The choice of a particular SIA approach depends on multiple considerations, such as the specific decision context, regional conditions, stakeholder preferences, capacities, data availability, budget and the time available for the assessment (König and others 2010). According to Bacic and others (2006), incomplete information is one of the main constraints on decision-making. To derive useful information for decision making, it is therefore necessary to understand causal linkages between human action and regional consequences (see, for example, Halvorson and others 2003; Hill and others 2006; Zhu and Dale 2000).

Much progress has been made in the study of complex human-nature relationships and interactions (see, for example, Bennett and others 2009; Liu and others 2007; Walmsley 2002). However, the scope of coverage of analytical and quantitative computer-based tools often remains limited due to predefined and tool-specific settings, particularly in ex ante studies (Uthes and others 2010), and because social considerations are still rare (Linkov and others 2006). This appears to be the case particularly in developing countries (Reidsma and others 2011), and new research strategies are required to understand the fundamental interactions between nature and society (Kates and others 2001; Lambin and others 2001). Sheate and others (2008) found that stakeholder participation can improve the understanding of causal linkages between land use and sustainability at the regional level. There is growing evidence that stakeholder participation can enhance decision-making processes (Reed 2008). Therefore, to promote responsible and transparent decision making, SIA requires participatory, structured and well-tested approaches that not only produce hard data but also integrate the knowledge of the different stakeholders and disciplines (Thabrew and others 2009; Zhen and others 2009).

Until now, most studies on soil and water conservation issues in the Oum Zessar watershed have focused on either environmental or economic aspects, for example, on the regional water balance (Schiettecatte and others 2005), the economic productivity of olive trees (Fleskens and others 2005) or the profitability of different water-harvesting systems (Ouessar and others 2004). Integrated impact assessments that consider the social, economic and environmental dimensions together are still missing.

This study was conducted as part of the EU funded LUPIS project (Land use Policies and Sustainable Development in Developing Countries, www.lupis.eu), which aims to develop integrated assessment tools in developing countries. The LUPIS project was built upon two larger EU projects: the SEAMLESS Integrated Framework (SEAMLESS-IF), concentrating on the agriculture sector across scales (van Ittersum and others 2008), and SENSOR, which focuses on multiple land use sectors at the regional scale (Helming and others 2011). Both approaches are indicator-based, integrate different quantitative and qualitative models and tools (Uthes and others 2010) and structure the assessment procedure according to the 'Driver-Pressure-State-Impact-Response' (DPSIR) framework of Smeets and Weterings (1999). The application of the FoPIA proposed here has been developed as part of the SENSOR project mainly to complement the development of computer-based SIA tools with

stakeholder knowledge, thereby allowing causal chains and assessment results to be verified from a regional perspective. However, the FoPIA could also be used to conduct a stand-alone impact assessment, for example, to study the effect of spatial planning policies on rural–urban sustainability in Indonesia (König and others 2010).

The first objective of our study was to conduct a stakeholder-based sustainability impact assessment of three alternative SWC scenarios in the Oum Zessar watershed. The FoPIA was adapted to the regional sustainability context and implemented to elaborate the alternative SWC scenarios and develop a set of regional assessment indicators, thereby assessing and exploring the possible impacts of the three SWC scenarios on sustainable development. Based on the results of the assessment, possible compromises and implications for regional land management and sustainable development are discussed. The second objective was to evaluate the applicability of the FoPIA method for impact assessment in the context of a developing country for the case of the Oum Zessar watershed in Tunisia.

Materials and Methods

The FoPIA Approach

Implementation of the FoPIA

The FoPIA approach provides a structured sequence of methods for conducting sustainability impact assessments of land use policies (König and others 2010; König and others in press; Morris and others 2011). The FoPIA consists of two assessment directions: firstly, a discursive examination of causal relationships and attributions of changes between human activities and sustainability targets, and secondly, the exploration of scenario impacts and possible trade-offs on defined sustainability targets at the regional level. The implementation structure of the FoPIA at the case study level has been adapted from Reidsma and others (2011) and follows three main steps: (i) scenario development, (ii) specification of the sustainability context, and (iii) scenario impact assessment (Fig. 1). The whole process is driven by regional data availability and the active participation of stakeholders in each assessment step. In addition to an extended preparation phase (mainly steps i and ii) to implement the FoPIA at the case study level and an assessment outcome evaluation phase, a two-day stakeholder workshop is at the core of this approach to scenario impact assessment.

Stakeholder Participation

Stakeholders are defined as individuals, groups, and organisations that are directly affected by decisions and actions or that have the power to influence the outcomes of these decisions (Freeman 1984). In this study, the stakeholder participation included: first, the consultation of individual local actors (mainly stakeholders and experts from regional land administration and research institutions) to support the development of regional SWC scenarios

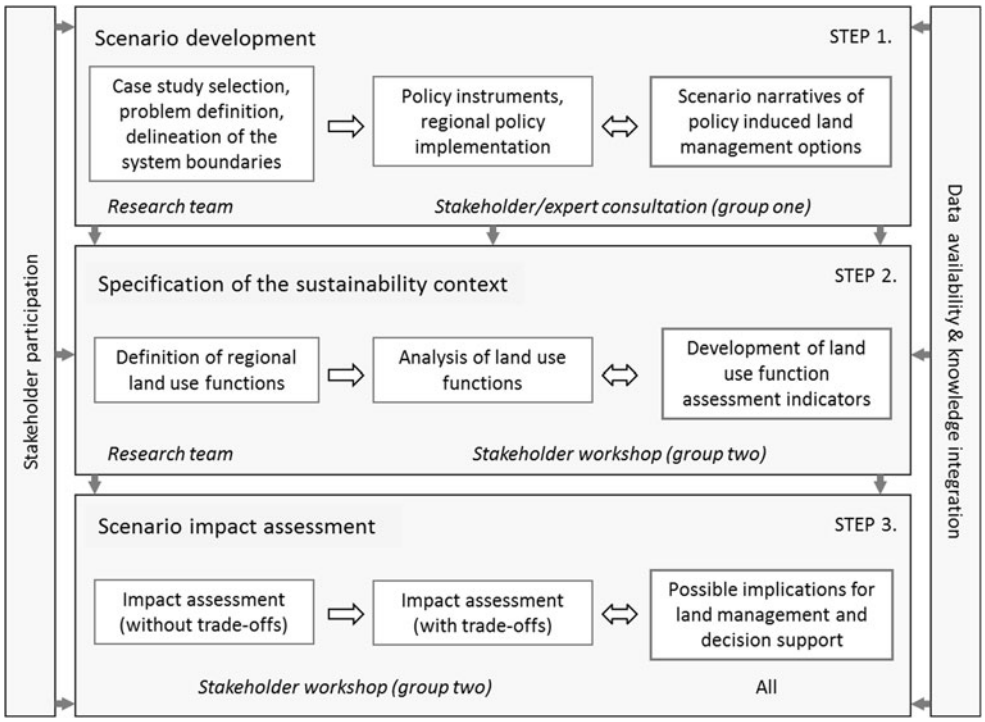

Fig. 1 The implementation structure of the Framework for Participatory Impact Assessment (FoPIA) to the case study of Oum Zessar watershed, Tunisia

(group one); second, a group of ten stakeholders was invited to perform the SWC scenario impact assessment, including regional authorities and administrators responsible for either the preparation of policy guidelines or regional policy implementation (group two).

The stakeholder selection criteria for the second group were: (i) the individual participant should have a strong background and knowledge about regional land use and SWC policy, and (ii) the group should be balanced among social, economic and environmental disciplines. The invited stakeholder group included six representatives from the Regional Administration of Agricultural Development (CRDA), two representatives of the South Development Office, Médenine (ODS), one expert of the Tunisian Union of Agriculture and Fishing (UTAP), and one expert of the Ministry of Environment and Sustainable Development (MEDD).

Scenario Development

Case Study Selection

The scenario development started with the selection of the case study region. For this study, the Oum Zessar watershed was selected as a case study of a region with an arid climate and low productivity land. The reason for selecting this case study was to assess the regional impacts of the SWC policy in terms of social, economic and environmental sustainability and to test the applicability of the FoPIA method in a North African country for the first time. The system boundaries were defined at the watershed level, which also represents an administrative level of policy implementation. The case study area is characterised by small-scale agriculture that is dominated by crop production, livestock farming, agro-forestry plantations with olive trees, and a growing sector of export-oriented date production. Water scarcity is considered the main limiting economic production factor for land, and rainfall varies between 150–230 mm rainfall per annum (Ouessar and others 2004). The population size was 24 188 in 1994 and reached 31 693 in 2008 (estimated based on INS[1] statistics 1994-2004), with an average growth rate of 2.3 % per annum.

Policy Instruments

As a response to high climate vulnerability in economic land production, the Tunisian government launched a policy on soil and water conservation (SWC) measures. The SWC policy started in the 1990 s through a two-decade strategy (1990–2000 and 2001–2011, Stratégies Décennales de CES) with the following goals (Ouessar and others 2009; Sghaier and others 2009):

- To protect arable lands against erosion, control water flux, increase land utilisation through the construction of water harvesting facilities, including (for example) field terraces in mountainous terrain (jessour system), large earthen dams (tabias), and groundwater recharge gabions (stone walls).
- To have farmers take responsibility for SWC with anti-erosion technologies according to the characteristics of their farms.
- To create private companies that specialise in SWC implementation and maintenance.

Scenario Narratives

Together with regional land administration experts and the Institut des Régions Arides, we developed a set of three alternative SWC scenarios for Oum Zessar watershed, including (i) the originally planned SWC implementation, (ii) the SWC actually implemented (in 2009) and (iii) a hypothetical expansion of SWC policy measures. We selected 2015 the target year for all scenarios, considering the following assumptions: the current implementation rate of SWC measures in 2009 was approximately 70 % of manageable land in the watershed and will not be changed until 2015. This is represented by the scenario of actual implementation denoted 'SWC-70'. Following the intended policy target, the originally planned SWC implementation will reach 85 % coverage of manageable land and is denoted the 'SWC-85' scenario. In addition to this, we also included a 'SWC-100' scenario, which represents a hypothetical expansion of SWC measures at full coverage of manageable land in the watershed. The latter scenario is currently under discussion and may be realised depending on the overall success, available budget and acceptance of this policy.

Specification of the Sustainability Context

Definition of Regional Land Use Functions

To capture the notion of sustainable development, the FoPIA makes use of the concept of land use functions (LUFs). LUFs are defined as 'goods and services', which are provided by different land uses that summarise the most relevant economic, social and environmental aspects of a region (Pérez-Soba and others 2008). LUFs build upon the concept of 'ecosystem services' but consider land use to be the main pressure for sustainability in rural areas, which can only be achieved if economic, social and environmental aspects are equally considered at the same level of investigation (Schößer and others 2010). In this study, a

[1] National Institute of Statistics, Tunisia.

generic set of nine social, economic and environmental LUFs as proposed in Pérez-Soba and others (2008) was used as the starting point to define a regional set of LUFs for Oum Zessar watershed.

Analysis of Land Use Functions

The regional set of LUFs was presented to a second group of ten invited stakeholders with different disciplinary backgrounds who were asked to assign weights of perceived regional importance to each LUF. The task was to assess differences in individual preferences and to invite stakeholders to enter a discussion on regional sustainability. A 0–10 scale was used to weight each LUF, where 0 indicates low importance and 10 very high importance. The same weight could be assigned to all LUFs. After a first scoring round, the average weights were presented back to the group. The ensuing discussions were used not only to clarify (and possibly adapt) definitions of LUFs, but also to reach a shared understanding of sustainability.

Development of Land Use Function Assessment Indicators

Each LUF needed to be represented by one associated assessment indicator in order to have a precise criterion for handling the LUF in the impact assessment. According to Walz (2000), an indicator is a variable that can be used to describe the state of a system and its changes. A generic set of LUF indicators developed for the context of a developing country (Indonesia) was adapted from König and others (2010) and presented to stakeholders in the second group in Oum Zessar watershed. The stakeholder group (group two) were given the opportunity to comment on and change indicators upon agreement among all group participants. This step was necessary to achieve a common understanding of the indicators, and it resulted in a commonly accepted set of LUF indicators. The following criteria were applied to specify the selection process:

1. the indicator should be relevant to the region and corresponding LUF
2. the indicator should be clear and understandable to all stakeholders
3. the indicator should be as precise as possible and measurable
4. the indicator should not be redundant, i.e., covered by another indicator.

Scenario Impact Assessment

Impact Assessment (Without Trade-Offs)

The scenario impact assessment was divided into two assessment parts: (i) a scenario impact assessment without trade-offs and (ii) a scenario impact assessment with trade-offs. The final outcome of the impact assessment was evaluated by the research team and considered in light of related studies and the available scientific literature. In the first step, a scoring scheme was used to assess both positive and negative impacts the SWC scenarios are anticipated to have on each LUF indicator using the following scoring scheme: 0 = no impact; −1 and +1 little impact; −2 and +2 high impact; and −3 and +3 extremely high impact. This scoring scheme entailed normalising the indicator values into one common scale. Upon completion of the individual scoring, average scores for the impact of each scenario on each LUF indicator were calculated (to one decimal point) and presented to the stakeholder group. To initiate a discussion, the moderator highlighted the contrasting positive and negative scores given by individual participants (scoring extremes) for each LUF and corresponding scenario. The stakeholder group was then asked to discuss the average scores and the various arguments behind the individual impact scoring results. Comments and individual impact arguments were recorded by the research team to underpin the assessment outcome with arguments and to note possible impact directions. After an exchange of arguments and open discussion, a second scoring round was undertaken in which group participants could adjust the scores of the first scoring round. The arguments were summarised and reported back by the moderator. The final results were adjusted upon common agreement of the group.

Impact Assessment (With Trade-Offs)

In order to assess impact trade-offs, the second stakeholder group assigned 'priority' rankings to all the LUFs, this time in a clear hierarchical order. A scoring scheme of 1–9 was applied, where 1 indicates lowest priority and 9 indicates highest priority. The impact assessment results (scores) and LUF ranking scores were aggregated along the three dimensions of sustainability (economic, social, environmental) using the following equation:

$$ri_d = \sum_{f=1}^{n} r_{f,d} * i_{f,d} \quad (1)$$

with ri = ranked impact, r = priorities assigned to each land use function (ranking), i = average impact on each land use function as assessed by the stakeholder group (impact assessment), d = sustainability dimension (economic, social, ecological), f = land use function (n = 9). The scenario impact scores and LUF ranks were aggregated to allow the impacts to be interpreted according to the priorities assigned to each LUF and thereby better understand the trade-offs associated with each scenario.

This allowed for the alternative scenarios to be compared based on which possible implications for land management and decision support are under consideration.

Results

Specification of the Sustainability Context

Definition and Analysis of Land Use Functions

The definition of the nine regionally adapted LUFs for the Oum Zessar watershed is presented in Table 1. The results of the regional importance analysis differed markedly among the economic, social and environmental land use functions in the Oum Zessar watershed as indicated by the averaged weights (Table 1). These revealed that primary production, the provision of work, quality of life, and the provision of abiotic resources are considered to be most important in the Oum Zessar watershed. Infrastructure, industry and services were given the lowest importance and cultural identity, biotic resources and ecosystem processes were scored of medium importance. This set of preferences seems plausible because primary production (mainly agricultural output) and related job opportunities (mainly on-farm work) have a long tradition in this region and still play a key role in rural regions of Tunisia (see Dhehibi and Lachaal 2006; Ouessar and others 2004). In contrast to this, stakeholders gave little weight to the LUF of 'industry and services' in this watershed because this area is remote and far away from industrial centres (Sghaier and others 2009). Similarly, infrastructure was not perceived to be of great importance for regional sustainability because most economic products are sold at local and regional markets. The weight attached to 'cultural identity', considered by the participants as preservation of agricultural heritage (i.e., traditional pastoralism, ancient water harvesting techniques and, recently, olive groves), plays an important role at this level and was suggested to be difficult to capture.

Changes in Land Use Function Preferences

After discussions and an exchange of arguments regarding the regional importance of LUFs, the changes in preferences adopted resulted in an increase of the average values for the environmental LUFs accompanied by increases in their economic dimension (Table 1). In particular, the two environmental LUFs, i.e., abiotic and biotic resources,

Table 1 Definition and analysis of land use functions in Oum Zessar watershed

	Land use functions (LUFs)	Regional relevance	Average score	Changes in scores (±)
Mainly social LUFs	*SOC 1—Provision of work*: provision of job opportunities for all in activities based on natural resources and related secondary sectors (e.g., agriculture, agro-forestry, processing industries)	Most people in the watershed rely on land-based job opportunities	7.9	0.0
	SOC 2—Quality of life: refers to an improved livelihood and a fulfilled minimum standard of living that enables a longer life expectancy	Life expectancy is a major attribute of improved quality of life	7.4	−0.2
	SOC 3—Cultural identity: refers to the maintenance and development of cultural heritage and identity	Cultural identity and traditions are very important to local people	6.0	−0.6
Mainly economic LUFs	*ECO 1—Industry and services:* including land-independent production systems, provision of space where residential, social and productive human activity takes place in a concentrated mode (e.g., settlements, industries and urban areas)	Built-up areas reflect economic development	5.1	+0.7
	ECO 2—Primary production: land that is used for production activities that do not result in irreversible change (e.g., crop- or forestland)	The majority of the local population is involved in agricultural production	8.0	+0.4
	ECO 3—Infrastructure: road networks, railways and public transport involving development that is largely irreversible	Transportation means link the region with the outside areas	4.4	+0.3
Mainly environmental LUFs	*ENV 1—Provision of abiotic resources*: the role of land in regulating the supply of water resources	Water is a basic natural resource	7.7	+1.7
	ENV 2—Provision of biotic resources: factors affecting the capacity of the land to support the provision of abiotic resources and ecosystem processes	Vegetation indicates an improved environment in arid regions	6.3	+1.6
	ENV 3—Maintenance of ecosystem processes: to ensure natural processes such as natural water balance, nutrient cycles and natural succession processes	Undisturbed land is less vulnerable to droughts and erosion	6.3	+0.4

were given higher scores than in the first round (+1.7 and +1.6), respectively. These changes were fostered by arguments brought forward by individual stakeholders stating that it is actually soil and water as natural resources that provide the basis for any primary production and related job opportunities. By contrast, the scores for culture and quality of life decreased slightly (-0.2, -0.6), based on the comments that (i) culture is still and will be a substantial part of the regional society and hence does not really require particular attention, and (ii) the quality of life has already experienced some improvements compared to the past but still remains of crucial importance.

Development of Land Use Function Assessment Indicators

The development and selection of assessment indicators was intensively discussed among the stakeholders, resulting in a commonly accepted set of nine land use function indicators (Table 2). There was more contention between stakeholders in relation to the selection of social indicators, whereas there was much less contention over the economic and (in particular) the environmental indicators.

Social Indicators

For the social dimension, employment rate was selected as the indicator for the LUF 'provision of work' because most people in the Oum Zessar watershed rely on regional job opportunities. Life expectancy was selected as the indicator for the LUF 'quality of life'. It was debated whether this indicator directly links to SWC policies. The indicator was finally accepted because life expectancy is a major attribute of improved quality of life and was expected to be affected by the promotion of SWC measures. For the LUF 'cultural identity', the proportion of traditional techniques used in land production was suggested to be the appropriate indicator. This indicator was accepted, but some stakeholders raised concerns about the feasibility of measuring this indicator.

Economic Indicators

The amount of regional financial investment resulting in new built-up areas was chosen as the indicator for the economic LUF 'industry and services'. Although the difficulty of measuring this indicator was mentioned by one stakeholder, the group agreed that this indicator is highly relevant to representing the overall economic development situation. For the LUF 'primary production', income per farmer was defined the indicator. The majority of the rural population generates their income from land-based activities, i.e., mainly from agricultural production such as market-oriented olive and date production. Road density was chosen as the indicator for the LUF 'infrastructure', given the fact that road transportation provides the link between rural areas and the rest of the region.

Environmental Indicators

For the environmental dimension, water availability per land holding was selected as the indicator for 'provision of abiotic resources.' This indicator was defined as 'water resource availability per land holding'. Permanent vegetation cover was selected as the indicator for 'provision of biotic resources', which is appropriate for measuring improvements of the overall environment in an arid region. Undisturbed land, lacking any anthropogenic activity, was chosen as the indicator for the LUF 'maintenance of ecosystem processes'. This indicator was widely accepted among the stakeholders because land becomes more vulnerable under certain land use management practices (for example, overgrazing due to high livestock density).

Impact Assessment of SWC Scenarios (Without Trade-offs)

In the first part, the results of the scenario impact assessment (without trade-offs) are shown in Table 3. A general observation from the results is that the assessed impacts

Table 2 LUF assessment indicators and stakeholders' acceptance

	LUF assessment indicator	Acceptance
SOC 1	Regional employment rate (%)	Widely accepted
SOC 2	Life expectancy (age)	Criticised but accepted
SOC 3	Traditional techniques (status)	Relevant but difficult to measure
ECO 1	Regional financial investments (td)	Accepted, but ambiguity was noted
ECO 2	Income per farmer (td)	Accepted
ECO 3	Road density (km length/region^{-1})	Accepted and specified
ENV 1	Water resource availability (m^3/land holding)	Widely accepted
ENV 2	Vegetation cover (area size/region^{-1})	Widely accepted
ENV 3	Natural land (area size/region^{-1})	Widely accepted

Table 3 Scenario impact assessment scores of three SWC policy scenarios

	SWC-70	SWC-85	SWC-100
SOC 1	0.3	1.5	1.9
SOC 2	0.4	0.7	0.9
SOC 3	1.5	0.8	0.5
ECO 1	0.4	0.6	0.3
ECO 2	1.3	1.6	2.0
ECO 3	0.0	0.5	1.4
ENV 1	0.4	1.7	2.4
ENV 2	0.7	0.4	0.1
ENV 3	−0.1	−0.5	−1.4

increased, both positively and negatively, in correlation with the increasing promotion of SWC measures.

Impact of the SWC-70 Scenario (Actual Policy Implementation)

The current SWC policy implementation, represented by the SWC-70 scenario, was assessed to have a positive impact on eight LUFs, particularly on the social LUF 'cultural identity' and the economic LUF 'primary production'. A negative impact was assessed for the environmental LUF 'maintenance of ecosystem processes'. Cultural identity was judged positive as a result of little impact on land management or the maintenance of traditional land management practices. Primary production was assessed positive because agriculture provides the main income source for rural people. However, the LUF 'maintenance of ecosystem processes' was judged to have a slightly negative impact due to land intensification in agriculture.

Impact of the SWC-85 Scenario (Intended Policy Implementation)

Full realisation of the originally planned SWC policy implementation was assessed to have positive impact on eight LUFs, particularly on the social LUF 'provision of work', the economic LUF 'primary production' and the environmental LUF 'provision of abiotic resources'. In this scenario, 'cultural identity' was assessed to perform less positively as a result of the impacts of land management changes. The impact on the environmental LUF 'maintenance of ecosystem processes' was again assessed to be negative. Provision of work and primary production were assessed to be positively affected because more land will be available for cultivation, providing more rural jobs and better economic return. The provision of abiotic resources, i.e., water availability, was judged to improve as a result of improved soil and water conservation activities. Again, the 'maintenance of ecosystem processes' was assessed to experience a negative impact due to increasing land intensification, particularly in the agricultural sector.

Impact of the SWC-100 Scenario (Hypothetical Expansion of SWC Policy Measures)

In the case where SWC policy measures are implemented at full coverage of manageable land in the watershed, the scenario was assessed to affect eight LUFs positively, with particular regard to the social LUF 'provision of work', the two economic LUFs 'primary production' and 'provision of infrastructure', and the environmental LUF 'provision of abiotic resources.' In addition to having an intensified impact on all LUFs compared to the SWC-70 and SWC-85 scenarios, this scenario was particularly expected to benefit infrastructure in terms of new road construction to access land that would otherwise stay out of production. Again, 'cultural identity' was assessed as declining compared to the SWC-70 and SWC-85 scenarios as a result of continued changes in land management. The environmental LUF 'maintenance of ecosystem processes' was assessed as experiencing significantly negative impacts due to further increases in land disturbance as a result of land use intensification.

Impact Assessment of SWC Scenarios (With Trade-Offs)

In this section, the results of the scenario impact assessment 'with trade-offs' are shown for each land use function (Table 4) and along the social, economic and environmental sustainability dimensions at higher aggregation (Table 5). At the aggregated level (Table 5), the promotion of SWC policy measures (SWC-85, SWC-100) was found to have benefits mainly on the social and economic dimensions. In the environmental dimension, the SWC-85 scenario, closely followed by the SWC-100 scenario, was assessed as providing the greatest benefit.

Table 4 Weighted scenario impacts for each land use function

	Ranking factor	SWC-70	SWC-85	SWC-100
SOC 1	0.60	0.18	0.90	1.14
SOC 2	0.56	0.23	0.39	0.51
SOC 3	0.49	0.73	0.39	0.24
ECO 1	0.36	0.15	0.22	0.11
ECO 2	0.63	0.81	1.00	1.25
ECO 3	0.36	0.00	0.18	0.51
ENV 1	0.56	0.23	0.96	1.35
ENV 2	0.55	0.39	0.22	0.06
ENV 3	0.39	−0.04	−0.19	−0.54

Table 5 Aggregated scenario impacts across the three dimensions of sustainability

Scenario	Social	Economic	Environmental
SWC-70	1.1	1.0	0.6
SWC-85	1.7	1.4	1.0
SWC-100	1.9	1.9	0.9

Social Impacts

Our results revealed that expansion of the SWC policy has the potential to improve some social functions in Oum Zessar watershed, particularly with regard to the provision of rural working opportunities (SOC 1) and an improved quality of life (SOC 2). The SWC's promotion of terraces for agricultural production, for example, enables the cultivation of land in areas that would otherwise stay out of production. The quality of life associated indicator 'life expectancy' was assessed to improve under the SWC policy based on the assumption that labour input per production unit could be reduced. Changes in traditional landscapes (SOC 3), however, were perceived as negative policy impacts.

Economic Impacts

The impacts of the policy on economic land use functions were assessed as positive, mainly due to primary production (ECO 2) and the provision of infrastructure (ECO 3). Although the promotion of terraces will provide land for economic production, new roads will also be needed to access these areas. Although the stakeholders reported a slight increase in the industry and service sector (ECO 1) as a result of economic development, a shift from primary production to off-farm work is not yet observed in the case of Oum Zessar.

Environmental Impacts

The impacts of SWC measures on environmental functions were primarily assessed as positive, particularly with regard to abiotic resources (ENV 1). Following the primary policy goal, the stakeholders judged that an expansion of SWC measures (SWC-85, SWC-100) would increase regional water availability in Oum Zessar watershed. However, negative impacts were assessed for the provision of natural vegetation (ENV 2) and some ecosystem processes (mainly the natural water cycle, ENV 3) due to anthropogenic activities and the disturbance of natural land.

Discussion

Sustainability impact assessment is an emerging field in the science-policy interface that meets the challenge of providing a 'holistic picture' of possible social, economic and environmental changes (George and Kirkpatrick 2006; Kates and others 2001; Pope and Grace 2006). In Europe, ex ante policy impact assessments have become a mandatory instrument (Tscherning and others 2008) due to the introduction of the EU impact assessment guidelines (Commission of the European Communities (CEC), 2005). In Europe, one main objective is to provide transparent information about possible policy effects and trade-offs and to improve the quality of policy decisions (De Smedt 2010). By contrast, in many developing countries policy assessments mainly focus on either environmental or economic aspects (see, for example, Baur and others 2003; Kirkpatrick and Parker 2004; Reidsma and others 2011). However, in both cases, a broad range of tools is available to conduct integrated assessments, although the balanced consideration of social, economic and environmental aspects is still hampered by the limitations of quantitative tools (Scrieciu 2007).

In the European context, advancements have been made in the analytical and integrated tools used to generate land use related policy assessments (see, for example, Sieber and others 2008; Uthes and others 2010; van Ittersum and others 2008). However, region- and tool-specific settings limit their "1:1" transferability to other regions where the input data and knowledge required to implement the tools is not available. The FoPIA method proposed here demonstrates how to conduct a holistic sustainability impact assessment of SWC policy by considering regional stakeholder knowledge as the main source of input information for the assessment.

Application of the FoPIA Approach to Oum Zessar Watershed

Our aim was to explore possible impacts of SWC policy on sustainable development in 2015 to support policy decisions in the near future. Although it is well known that the policy effects may take a long time on the ground, stakeholders were able to share their initial experiences with the SWC policy and discuss principle insights regarding causal linkages between regional policy-land-sustainability relationships. In general, the SWC policy was assessed to have achieved its main goals of enhancing the capacity of the land for production. The introduction of field terraces and other water-harvesting techniques enables the use of land in arid areas that would otherwise stay out of production (Amsalu and de Graaff 2007; Fleskens and others 2005) and also reduces the risk of harvest losses during dry years while locally stabilising the groundwater level (Schiettecatte and others 2005). According to Hill and Woodland (2003), these jessour systems are also difficult and costly to maintain while frequently suffer sedimentation problems.

At the national level, agricultural productivity increases an average of three percent annually (Dhehibi and Lachaal 2006), which indicates the potential for economic land production. The use of water-harvesting techniques in southern Tunisia has a long tradition (Hill and Woodland 2003), and farmers are well equipped to produce food and survive in such harsh environmental conditions. However, with increasing demand to also use land for market and export-oriented production (mainly olives and dates), the traditional system will likely change towards larger-scale and commercialised practices. In this regard, Hill and Woodland (2003) suggested combining the modern measures of the large-scale SWC program with traditional practices, an approach that could potentially reduce negative policy effects such as small-scale farmers being neglected by program benefits (see Brismar 2002), causing disadvantages in competiveness.

Our study revealed that SWC-induced 'land modifications' entail changes in socio-environmental functions that are partly perceived negatively by regional stakeholders. At this stage, policy makers have to evaluate the potential trade-offs between policy goals, stakeholder preferences and potential policy risks. It could be argued that modernisation and development will always entail changes in regional traditions. However, it is also the responsibility of decision makers to communicate these potential changes to local stakeholders in order to achieve greater acceptance and fuller implementation of the policy interventions. An importance assessment revealed the potential risk of SWC having unintended effects such as overexploitation of natural resources as a result of overgrazing. Livestock keeping and nomadic pastoralism have a long tradition in southern Tunisia (H.Steinfeld and others 2010). However, competition between land users and increasing numbers of livestock have contributed to severe land degradation problems in this region (Jeddi and Chaieb 2010).

Some Methodological Reflections on the FoPIA Methodology

The FoPIA approach was critically evaluated and discussed with the local research team (mainly in terms of methodological potential and limitations) and with the stakeholder group (mainly in terms of the potential utility of the outcome). The quantitative information required for scenario development was derived from the literature, whereas local knowledge was a crucial complement to the implicit information for the regional SWC policy implementation. Similar to a study conducted by Holman and others (2008), we found that stakeholder participation helped to elaborate policy scenario assumptions and to achieve high acceptance for the scenarios. The indicator 'development' appeared to be the most challenging because all three dimensions of sustainability had to be considered in its assessment. On the one hand, the stakeholders had different disciplinary backgrounds, and on the other hand, an interdisciplinary set of social, economic and environmental assessment indicators had to be developed in a balanced way that was acceptable to the whole stakeholder group. In particular, the social indicators needed to be extensively discussed because they require a more qualitative type of consideration that, in general, makes them more difficult to capture (see Singh and others 2009; Slee 2007). The land use functions helped to ensure a balanced development of indicators and to forge links between the SWC policy, regional land management, and sustainability. Aggregation of the assessment results, also based on the LUF approach, provided an estimate of the contributions of individual scenarios to sustainability, from which implications for regional land management could be drawn. The impact assessment considered those stakeholders who were responsible for the regional planning, support and/or implementation of the SWC policy. The invited stakeholder group (group two) had a strong regional background and was able to follow the assessment procedure. Although an attempt was made to balance the group among disciplines, it appeared that the participants had stronger backgrounds in economic and environmental aspects. The stakeholder group positively rated the opportunity to jointly assess and explore possible SWC consequences in a structured and research-based setting. The opportunity to clarify causal relationships between policy actions and regional sustainability and to reflect on and learn from discussions and illustrations of assessment results was very much appreciated by the participants. We conclude that the FoPIA could be a useful tool, for example, for researchers or consultancies to support the implementation of regional sustainable development strategies.

Implications for Regional SWC Policy Implementation

Based on the assessment results, full implementation of the SWC (SWC-100) could potentially have the highest contribution to sustainability, and implementation at 85 % (SWC-85) was predicted to have comparable effects, albeit with less serious negative impacts on ecosystem processes. Considering the fact that the full implementation of SWC measures (SWC-100) also entails negative impacts on the natural system, SWC implementation at 85 % might be preferable.

Conclusions

In this study, we demonstrated how to implement and conduct an impact assessment of alternative SWC policy

scenarios in the Oum Zessar watershed, southern Tunisia, using the participatory FoPIA assessment tool. A main outcome of applying the FoPIA is the learning process during the assessment procedure. This process raises awareness of the possible consequences of policy decisions in the light of sustainable development while also informing key stakeholders involved in decision making about possible directions for policy improvement. Our study showed a differential impact of SWC scenarios on selected social, economic and environmental land use functions:

- The results of the impact assessment showed that the selected social and economic land use functions mainly benefit under the promotion of SWC measures as a result of enhanced water resource availability for land-based production.
- The land use functions assessed to be of the highest important by the stakeholders were also targeted by SWC policies, i.e., the provision of abiotic resources (water availability, ENV 1), primary production (ECO 2), and the provision of work (SOC 1).
- The LUF 'Cultural identity' was assessed as having a negative impact if SWC measures are expanded due to changes in 'traditional (SCO 3)' landscapes (e.g., introduction of dams, gabions, etc.).
- The expansion of SWC measures in the Oum Zessar watershed was assessed to have negative impacts on the natural vegetation (ENV 2) and key ecosystem processes (ENV 3) as a result of changes in land management accompanied by increasing land use pressure.

Based on the assessment results, an implementation of the SWC policy at 100 % coverage of manageable land could be suggested if achieving the maximum socio-economic benefit is the goal. However, if stakeholders' preferences are taken into account, and considering the fact that the implementation of SWC measures also implies some negative changes to the natural system, SWC implementation at 85 % might be preferable.

From a methodological perspective, the role of stakeholder participation appeared to be crucial during all assessment steps for the consideration of region-specific and implicit information:

- to translate policy measures into a plausible set of regional land management scenarios,
- to analyse the sustainability context and key land use functions,
- to develop operational and regionally relevant assessment indicators,
- to assess and explore SWC scenario impacts and trade-offs, and
- to derive possible suggestions for implications in land management.

We conclude that participatory impact assessment contributes to an enhanced regional understanding of key linkages between policy effects and sustainable development. On the one hand, stakeholder knowledge was used to provide a region-specific and holistic picture of the potential impacts of policy on sustainable development. On the other hand, in addition to providing their knowledge, the stakeholders also benefited from the impact assessment process. Land administrators and authorities have an interest in successful and efficient policy implementation. Active participation during the impact assessment provides a platform for social learning, exchange and better understanding of the complexity of human-nature interactions. We believe that a better understanding of this complexity will translate into sustainable and efficient policy implementation at the regional level.

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